Data Engineering Integration 10.4.0: Security for AWS Databricks

In this article, learn the security functionalities that Data Engineering Integration supports and how to configure them in an AWS Databricks environment.

Overview

Informatica Data Engineering Integration provides a holistic support for the security layers of the Databricks cluster in order to keep your data and your enterprise secure. To integrate Data Engineering Integration with a secure Databricks cluster, it is not mandatory to configure security on the Informatica domain.

This article discusses Data Engineering Integration support for authentication, authorization, and encryption mechanisms that an AWS Databricks cluster uses.

You can apply the following types of encryption to data in the AWS Databricks environment:

- At-rest **SSE-S3** encryption. When you use server-side encryption (SSE) of S3 data, you request Amazon S3 to encrypt data, using a master key, before writing it to S3 buckets and decrypting it when you read from S3. See AWS documentation.
- At-rest **SSE-KMS** encryption. Use customer master keys (CMK) for authorization when encrypting and decrypting data in S3 buckets. See AWS documentation.
- At-rest **encryption of local disks using KMS.** Encryption of data stored on local disks using a KMS customer-managed key (CMK). See AWS documentation.

For information about integrating Data Engineering Integration with the AWS platform, refer to the Data Engineering Integration Guide.

Supported Security Systems for AWS Databricks Clusters

Encryption and Authentication

The following table lists encryption and authentication methods in the Databricks environment:

Encryption Type	Encryption	Authentication
At-rest encryption of data in S3	• SSE-S3 • SSE-KMS	IAM roles [1]
At-rest encryption of data on local disks	• KMS	N/A

[1] See the Databricks documentation.

Authorization

You can mount Databricks file system (DBFS) directories and access object storage in those directories without enabling additional access. To enable DBFS access, simply ensure that users have read and write acess. See the Databricks documentation.

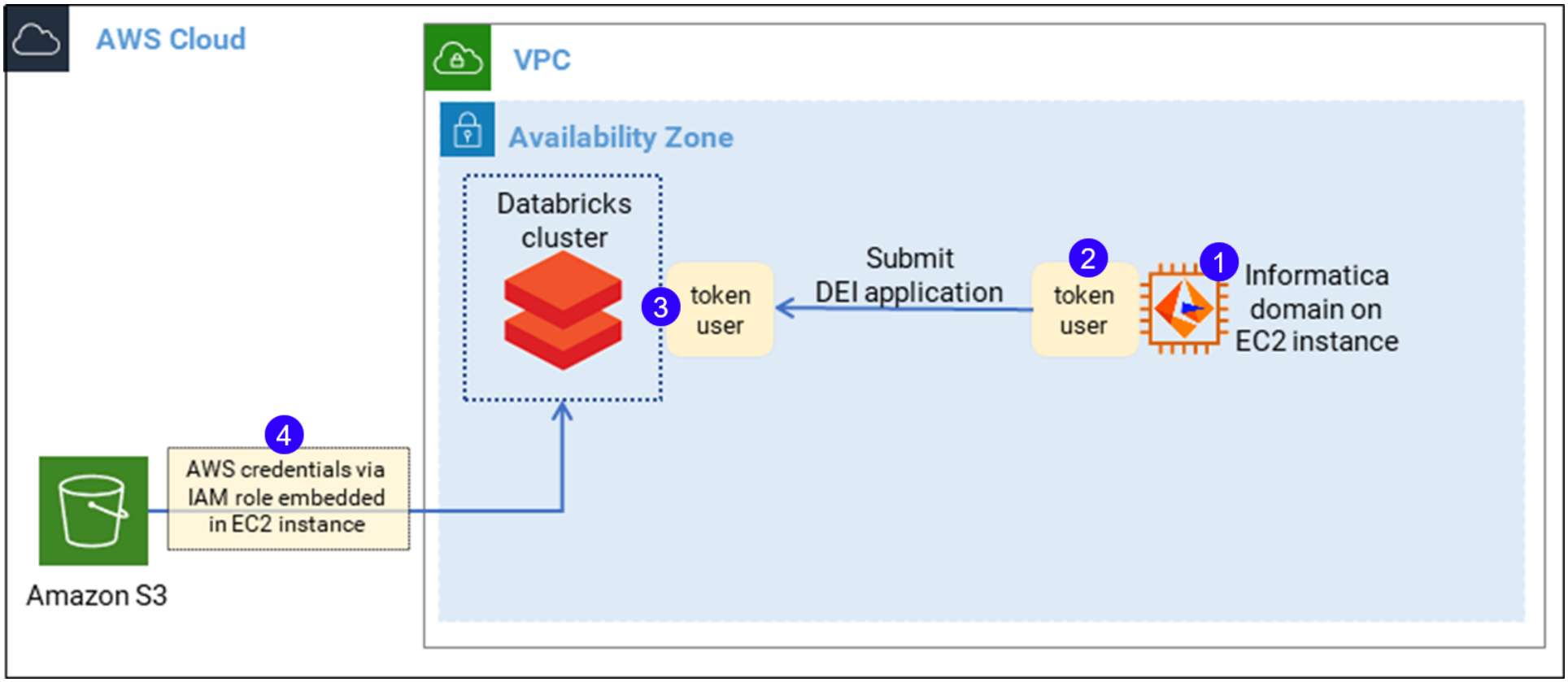
Supported Informatica Adapters for AWS Databricks Clusters

The S3 adapter supports at-rest encryption for SSE-S3, SSE-KMS, and local disk encryption. For more information, see the PowerExchange Adapter for S3 User Guide and the Amazon S3 Developer Guide.

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Security Configuration for Databricks Clusters and S3 Resources on AWS

The following image shows security configuration options for a deployment of Data Engineering Integration on AWS integrated with a Databricks cluster.



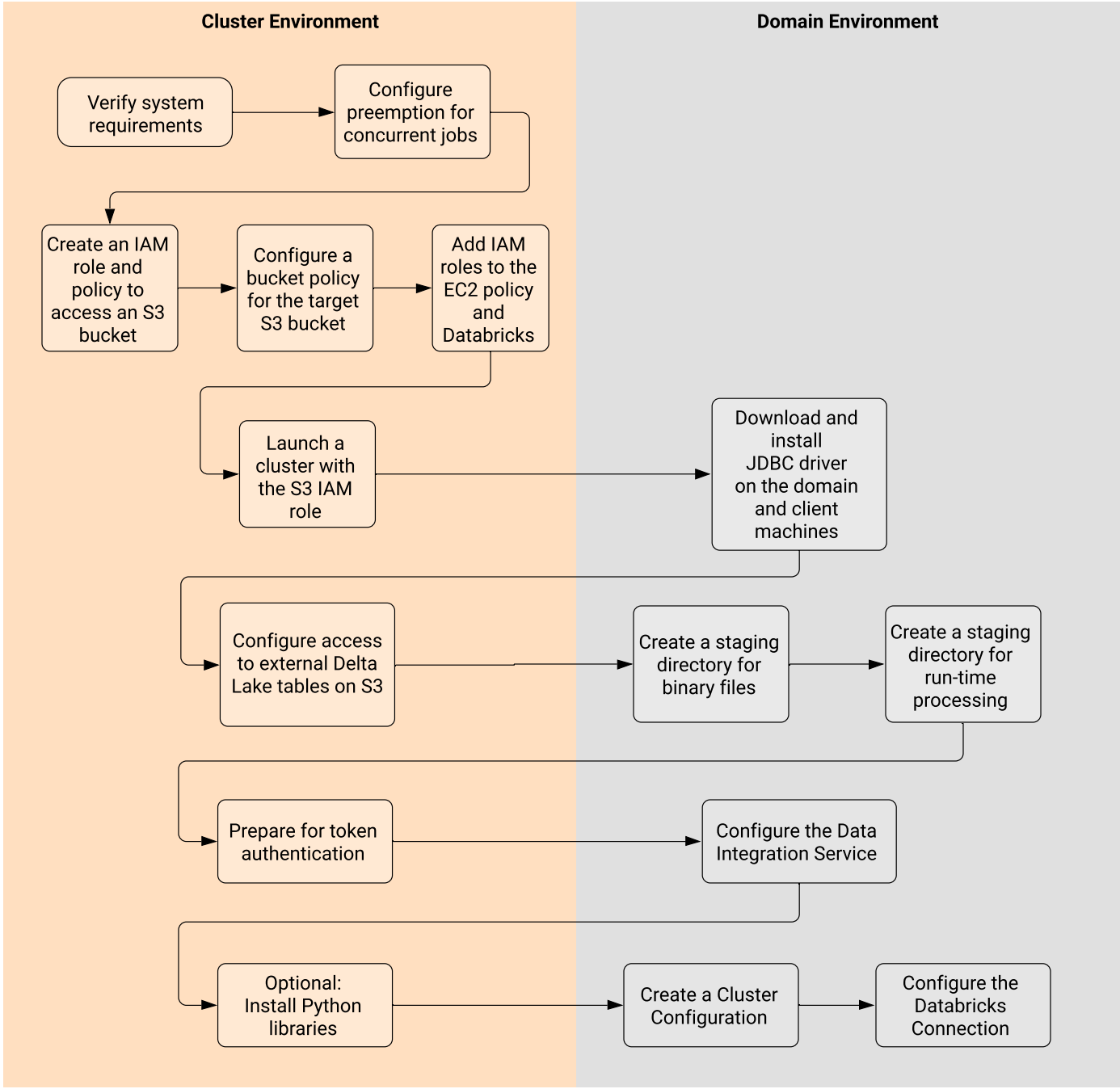
The following list of elements matches the numbered elements in the image:

1. The Informatica domain is deployed on an EC2 instance in an AWS VPC.
2. The Data Integration Service submits a job to the cluster as the user associated with the authetication token.
3. The Databricks cluster executes jobs as the user associated with the authetication token.
4. The Databricks cluster accesses S3 resources using credentials certified by IAM roles set up for the cluster in the AWS console.

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Security Configuration for AWS Databricks Using Delta Lake Resources in S3

The following flowchart shows the Data Engineering Integration security configuration tasks and where to perform the tasks. Click on the task to see the topic in documentation.



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